When and Why Electricity Prices Go Negative: An Empirical Study of CAISO and MISO, 2023–2024

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June 9, 2026

Abstract

We study day-ahead negative locational marginal prices (LMPs) in CAISO and MISO during 2023–2024. Negative prices in CAISO concentrate around midday in spring, while negative prices at selected MISO wind-belt nodes occur throughout the day. A two-stage classifier–regressor improves forecast accuracy in CAISO but performs worse than an end-to-end LightGBM model in MISO. SHAP attribution identifies the first-stage output as the source of the MISO error. Binarizing this output before passing it to the regression stage improves MISO accuracy relative to both the end-to-end baseline and the original continuous handoff. Battery-arbitrage backtests show that the resulting forecasts also improve operational value: forecast-driven dispatch captures approximately 96% of perfect-foresight revenue in CAISO, and the binary handoff increases MISO revenue by $1,019/MW-yr relative to the continuous handoff. We also evaluate distributional forecasts. Split-conformal calibration improves prediction-interval coverage but does not increase arbitrage revenue.

1 Introduction

1.1 Locational Marginal Prices

In wholesale electricity market, the locational marginal price (LMP) is the marginal cost of serving one additional megawatt-hour (MWh) of demand at a specific location and time. CAISO (the California Independent System Operator) and MISO (the Midcontinent ISO) operate day-ahead and real-time markets. This study focuses on hourly day-ahead LMPs. Although LMPs are usually positive, they can be negative, requiring generators to compensate consumers for incremental consumption.

1.2 Drivers of Negative Prices

Three mechanisms drive negative LMPs in U.S. markets:

- **Renewable oversupply.** Solar and wind generation has near-zero marginal cost. When supply exceeds short-run demand, prices may clear at or below zero.
- **Production Tax Credit.** Eligible generators receive a federal subsidy for each MWh they produce, so they may offer electricity at negative prices and still earn positive net revenue. When these negative offers set the market-clearing price, the LMP falls below zero. Consumers are then paid to use electricity during that interval. The current credit level can support prices of approximately −$25 to −$30/MWh.
- **Transmission congestion.** Binding transmission constraints can produce negative local prices even when the broader system price is positive. This mechanism is particularly relevant in MISO.

Negative prices affect battery operators, renewable generators, and system planners. Their frequency and magnitude influence storage dispatch, generation revenue, transmission planning, and demand-response design.

1.3 Research Questions

RQ1. When and where do negative LMPs concentrate in CAISO and MISO?

RQ2. How accurately can machine-learning models forecast next-day negative-LMP hours from publicly available data, and how does model architecture affect performance?

RQ3. How much operational value do these forecasts provide in a battery-arbitrage backtest?

Section 4 describes the AutoResearch workflow under which the study was conducted. Section 5 presents the empirical results, and Section 6 discusses their implications. Other technical details are documented in the appendices.

1.4 Contributions

This paper makes four contributions:

1. We characterize two distinct negative-price regimes: a concentrated midday pattern in CAISO and a more diffuse pattern at selected MISO wind-belt nodes.
2. We show that the two-stage forecasting model improves accuracy in CAISO but performs worse in MISO. SHAP attribution indicates that the MISO regression stage misuses the first-stage probability as a proxy for price magnitude.
3. We propose a simple modification: convert the first-stage probability into a binary indicator before passing it to the regression stage. This improves MISO forecasting performance.
4. We evaluate the forecasts at the operational level with a battery-arbitrage backtest. The modified model increases realized arbitrage revenue in MISO, while conformal prediction improves uncertainty calibration but does not increase revenue under symmetric arbitrage.

2 Related Work

Electricity-price forecasting and negative-price events. Day-ahead electricity-price forecasting is a mature literature with established benchmark models, rolling-origin evaluation protocols, and statistical comparisons of predictive accuracy [1]. Negative prices are a distinct forecasting challenge because they occur intermittently and arise from multiple market mechanisms, including renewable oversupply, production incentives, and transmission congestion [2–5]. A relevant modeling strategy is the two-stage spike–regression decomposition [6], which first estimates the probability of an extreme-price event and then predicts the price level using that probability as an input. The original application focuses on positive price spikes; we adapt the same structure to negative prices. To evaluate the operational significance of forecast improvements, we follow the storage-operations literature and translate price forecasts into realized revenue through a battery-arbitrage linear-program backtest [7, 8].

The two-stage decomposition is not the only approach to electricity-price forecasting. A common alternative is to predict the price directly with a single model, using regularized linear models such as LEAR, tree-based models, or neural networks [1]. Probabilistic approaches instead estimate conditional quantiles or full predictive distributions, allowing the forecaster to represent uncertainty explicitly [9]. In this paper, the end-to-end LightGBM baseline represents the direct point-forecasting approach, while the quantile-regression extension represents the probabilistic approach.

Statistical methods. Several statistical tools support the empirical analysis. The Diebold–Mariano test [10], with the Harvey–Leybourne–Newbold small-sample correction [11], evaluates whether differences in forecast accuracy are statistically significant. SHAP attribution measures the contribution of each feature to an individual model prediction [12, 13]. Quantile regression estimates conditional prediction intervals [14], and Conformalized Quantile Regression (CQR) adjusts those intervals to achieve marginal coverage [15–17].

Data sources. The empirical analysis uses public day-ahead LMP data from CAISO OASIS [18, 19] and MISO Data Exchange [20], accessed through the `gridstatus` Python library [21].

CAISO market-monitoring reports and MISO market-monitoring reports provide institutional context for the observed negative-price regimes [2, 3].

3 Methodology

3.1 Forecasting Models

We adapt a two-stage forecaster for price-spike prediction [6]. A LightGBM classifier estimates the probability of a negative LMP, and a LightGBM regressor predicts the LMP using that probability as an additional feature [22]. We evaluate both a continuous-probability handoff and a binarized handoff.

The baselines are a seasonal-naive forecast using the previous day's LMP at the same hour, an end-to-end LightGBM regressor, and a perfect-foresight oracle. Appendix A provides the full specification.

3.2 Data

We use day-ahead hourly LMPs from CAISO and MISO during 2023–2024, accessed through `gridstatus` [21]. The sample includes three CAISO trading hubs and four MISO wind-belt nodes. Named MISO hubs are excluded because aggregation produces near-zero negative-price rates. Training, validation, and test splits are chronological; the spring shoulder season is used for validation and threshold tuning. Appendix C provides the full data specification.

3.3 Evaluation

Classification metrics are PR-AUC, recall, and Brier score. Regression metrics are overall MAE and MAE on negative-price observations. Pairwise accuracy comparisons use the Diebold–Mariano test [10] with the Harvey–Leybourne–Newbold correction [11]. Appendix B provides the full definitions.

Operational value is measured with a battery-arbitrage backtest [7, 8]. We simulate a grid-connected battery with 4 MWh of storage capacity and a 1 MW maximum charge or discharge rate. For each day, the battery chooses when to charge from the grid and when to sell electricity back to the grid based on forecast prices. The resulting schedule is settled at realized prices, and revenue is reported on an annualized per-MW basis. Appendix D provides the full formulation.

4 AutoResearch Workflow

The empirical study reported here was conducted by an AI agent under explicit human checkpoints. The human authors supplied the research specification, provisioned the data-source credentials, and reviewed the final artifact; the agent owned every step between those checkpoints. This section describes the two execution loops that distinguish "AutoResearch" from a one-shot pipeline, the harness mechanism through which the agent accumulates constraints from observed failures, and the agent's responsibilities within each iteration.

4.1 Execution loops

The pipeline contains two distinct feedback loops, illustrated in Figure 1.

- **Implementation loop** (inner). If verification fails because of a code, data, or numerical issue, such as a missing column in an ISO response, a leakage between train and test, an unstable training run, or a misaligned timestamp, the agent returns to *Implement* without revising the hypotheses. The agent iterates on the code itself, with each iteration committed under version control for audit.
- **Hypothesis loop** (outer). If verification passes mechanically but the results indicate the hypotheses themselves require revision, the agent returns to *Plan and hypothesize*, logs the revision, and proceeds with the updated hypothesis. The original and revised hypotheses are both retained in the final report so that the research process is auditable. In the present study,

the hypothesis loop fired once during the MISO Stage-2 analysis: the initial expectation that the two-stage decomposition would improve accuracy in both markets was contradicted by the MISO MAE result of §5.2, which triggered the SHAP investigation of §5.3 and the binarized-handoff intervention of §5.4.

4.2 Harness: accumulating constraints from failure

Every iteration that triggers a return to *Implement* or *Plan* also produces a structured *failure-mode entry*: what was attempted, what verification surfaced, and the corrective action. After each phase the agent distills these entries into a unified *harness*, a living artifact that combines the original human specification with the invariants the agent must honor in subsequent iterations. Concrete examples of invariants written to the harness during the present study include: "all DA features must be timestamped no later than $D-1$ at the relevant gate-closure time" (from a leakage failure during feature construction); "Stage-1 class weights must be set by inverse base rate rather than 1:1" (from a Stage-1 PR-AUC underperformance); and "cross-market architecture comparisons must be reported with paired-bootstrap confidence intervals on per-(location, day) realized revenue" (from the battery-arbitrage backtest evaluation).

The harness is the runtime control surface for the agent. At every planning step the agent reads the current harness and produces a plan that respects each invariant in it; the verify stage enforces both the original spec and the accumulated invariants. By design the harness is monotone: entries are added only when justified by an observed failure, and are removed only by explicit human review. The agent's effective behavior therefore tightens monotonically across the project, and the harness itself is checked into the iteration log alongside the artifacts of §4.

5 Results

This section presents the empirical results in the order of the research questions.

5.1 Spatial and Temporal Distribution of Negative Prices

The CAISO solar-driven midday concentration in the spring shoulder season and the diffuse 24-hour pattern at MISO wind-belt locations are documented qualitatively in market-monitoring reports [2–4]. Figure 2 quantifies these patterns at the selected locations during 2023–2024. The named MISO hubs have approximately zero negative-price observations because aggregation masks local price variation; Appendix C documents the selected MISO nodes.

The two heat maps exhibit distinct structures. CAISO has a concentrated midday region during April and May, whereas MISO has bands that extend across the day with seasonal peaks in spring and autumn. We therefore report forecasting results separately for CAISO and MISO.

5.2 Negative-Price Classification and Price Regression

The Stage-1 classifier identifies negative-price hours in both markets. On the test set, PR-AUC is 0.890 in CAISO and 0.923 in MISO, compared with 0.261 and 0.257 for the seasonal-naive baseline. Recall exceeds 0.80 in both markets.

Table 1: Stage-1 classification performance on the test set.

	CAISO	**MISO**
Stage-1 PR-AUC	0.890	0.923
Seasonal-naive PR-AUC on test	0.261	0.257
Ratio to seasonal naive	3.4×	3.6×
Recall at tuned threshold	0.81	0.82

The Stage-2 regression results differ by market. Relative to the end-to-end LightGBM baseline, the two-stage model reduces MAE on negative-price observations from \$4.47/MWh to \$4.07/MWh in CAISO ($p \approx 1.7 \times 10^{-5}$), but increases MAE from \$8.08/MWh to \$8.54/MWh in MISO ($p \approx 4.2 \times 10^{-13}$).

The markets therefore differ not only in the distribution of negative prices, as shown in Section 5.1, but also in how predictable those prices are under the two-stage model. The next sections identify the source of the MISO error and modify the model accordingly.

5.3 SHAP Attribution of the Cross-Market Difference

SHAP attribution [12, 13] explains the MISO result. A SHAP value measures how much a feature changes an individual prediction. Figure 3 reports the mean absolute SHAP value of each Stage-2 feature; a larger value indicates a more important feature. The higher SHAP value of `p_hat` in MISO does not by itself explain the performance gap. We therefore compare how `p_hat` enters the Stage-2 prediction in the two markets.

Figure 4 shows a clear difference. In CAISO, `p_hat` is nearly binary. In MISO, it varies continuously, and larger values are associated with lower Stage-2 predictions. This suggests a simple test: replace the continuous MISO probability with a binary approximation and evaluate whether performance improves.

5.4 Binarized Stage-1 Handoff

We replace `p_hat` with a binary indicator before passing it to Stage 2.

Table 2: MISO MAE on negative-price observations. "Δ vs single-LGBM" is relative to the end-to-end LightGBM baseline; negative values indicate lower error.

	MAE-neg, \$/MWh	**Δ vs single-LGBM**	**DM p-value**
Two-stage, continuous `p_hat` (original)	8.537	+0.46 (worse)	4.2×10^{-13}
Two-stage, binarized `p_hat`	**7.925**	−0.16 **(better)**	4.0×10^{-6}
End-to-end single-LGBM	8.080	—	—

The binary handoff reduces MISO MAE from 8.537 to 7.925 \$/MWh and outperforms the end-to-end baseline. This result supports the SHAP-based explanation.

5.5 Operational Value in Battery Arbitrage

We measure operational value with the battery-arbitrage backtest defined in Appendix D. The battery schedules charging and discharging using forecast prices; revenue is calculated using realized prices.

In CAISO, all forecasting models earn approximately \$53,000–55,000/MW-yr. In MISO, model choice matters:

Table 3: MISO annualized battery-arbitrage revenue. "Pct of oracle" is the share of perfect-foresight revenue. "Δ vs end-to-end" is the difference from the end-to-end LightGBM baseline.

Model	**\$/MW-yr**	**Pct of oracle**	**Δ vs end-to-end**	**Paired p**
Oracle (perfect foresight)	47,298	100.0%	—	—
Two-stage, binarized `p_hat`	**36,602**	**77.4%**	**+\$243**	**0.058**
End-to-end single-LGBM	36,359	76.9%	0	—
Two-stage, continuous `p_hat`	35,582	75.2%	−\$777	≈ 0
Seasonal-naive	20,020	42.3%	−\$16,339	≈ 0

The binary handoff increases MISO revenue by \$1,019/MW-yr relative to the continuous handoff and by \$243/MW-yr relative to the end-to-end baseline ($p = 0.058$). The seasonal-naive baseline captures only 42% of perfect-foresight revenue.

5.6 Prediction Intervals and Risk-Aware Dispatch

The previous analysis uses a single predicted price for each hour. We next test whether a range of plausible prices can improve battery dispatch. A battery operator could use this range to avoid charging or discharging when the price forecast is too uncertain.

We estimate price ranges with quantile regression [14]. For example, the predicted P10–P90 interval is intended to contain the realized price 80% of the time. However, the raw intervals are too narrow: the nominal 80% and 90% intervals contain only 59% and 74% of realized prices.

We correct the intervals with split-conformal calibration [16, 17]. This procedure measures the interval errors on a held-out calibration period and widens future intervals accordingly. Appendix F provides the calibration rule. After calibration, the empirical coverage is close to the intended coverage:

Figure 7 compares the following dispatch policies:

- **L1 binarized**: the point-forecast model from Section 5.5.
- **P50 dispatch**: optimize using the median quantile forecast.
- **Robust LP (naive 80% PI)**: assume the upper end of the raw 80% interval when charging and the lower end when discharging.
- **Robust LP (conformal 90% PI)**: apply the same rule using the calibrated 90% interval.
- **CC dispatch**: charge only when the full interval is below a buy threshold, and discharge only when the full interval is above a sell threshold.
- **Pessimistic-P05 dispatch**: optimize using the lower quantile forecast.

Appendix G provides the full policy definitions.

None of the interval-based policies outperforms the point-forecast baseline. Calibration makes the price ranges more reliable, but conservative dispatch reduces arbitrage revenue. Prediction intervals may still be useful when the operator faces asymmetric costs, such as reliability penalties.

6 Discussion

6.1 Forecasting Architecture

- **Architecture details matter across regimes.** The same two-stage forecaster that improves accuracy in CAISO underperforms in MISO unless the Stage-1 probability is binarized before the Stage-2 handoff. Cross-market validation is therefore important when selecting an architecture.
- **The interface between stages should reflect its intended interpretation.** A binary regime gate prevents Stage 2 from treating the first-stage probability as a continuous magnitude signal.
- **Feature-dependence plots provide a useful diagnostic.** The cross-market contrast in how Stage 2 uses the Stage-1 probability (Figure 4) identifies the source of the MISO error.

6.2 Battery Dispatch

- **The value of improved forecasts depends on regime complexity.** In CAISO, machine-learning forecasts barely outperform the seasonal-naive baseline. In MISO, machine learning captures 77% of oracle revenue compared with 42% for the seasonal-naive baseline.
- **Calibrated intervals are most relevant to asymmetric decisions.** Under the symmetric arbitrage objective tested here, distributional forecasts do not improve revenue. Their operational value should be evaluated in settings with reliability penalties, hedging costs, or ancillary-service obligations.

6.3 Implications for Electricity-Price Forecasting

- **Single-market evaluations can produce regime-specific conclusions.** Architectural failure modes may surface only across qualitatively different markets.

- **Diebold–Mariano comparisons can reverse across regimes.** The continuous-probability architecture improves accuracy in CAISO and reduces it in MISO.
- **The relationship between MAE and realized arbitrage revenue is regime-dependent.** Statistical improvements in forecast accuracy should be paired with operational backtests.
- **Distributional forecasts and point forecasts address distinct objectives.** Calibrated uncertainty is most valuable when the downstream decision problem includes asymmetric losses.

7 Limitations and future work

We list briefly the principal caveats; the longer list is in Appendix H.

- **Two years, two ISOs.** CAISO and MISO span renewable-driven and wind/congestion-driven regimes, but a third ISO such as ERCOT is needed to evaluate whether the observed distinction generalizes.
- **Pre-2023 CAISO data are unavailable through the selected source.** The empirical split is 2023-03 to 2024-02 for training, 2024-03 to 2024-05 for validation, and 2024-06 to 2024-12 for testing.
- **Weather features were not included.** NSRDB irradiance and WIND Toolkit wind-speed data require NREL API credentials that were unavailable for this study.
- **The battery backtest uses daily SoC reset and a single battery configuration.** Real operators carry SoC across days and might solve a 7-day LP. Relaxing the daily reset would raise absolute revenue numbers but is unlikely to change the model ranking on a per-day basis.
- **Risk-aware dispatch was tested under a single decision problem.** Future work should evaluate settings with asymmetric losses, such as reliability contracts or hedging.

8 Conclusion

Negative-price regimes differ materially across CAISO and MISO. A continuous-probability two-stage architecture improves MAE on negative-price observations in CAISO but underperforms in MISO because the regression stage uses the first-stage probability as a magnitude proxy. Binarizing this handoff improves MISO accuracy and adds approximately $1,000/MW-yr in realized battery-arbitrage revenue relative to the continuous architecture. Conformal calibration restores prediction-interval coverage but does not improve symmetric arbitrage under the tested policies. These findings support cross-market model evaluation and the use of operational backtests alongside statistical forecast metrics.

9 References

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Figures

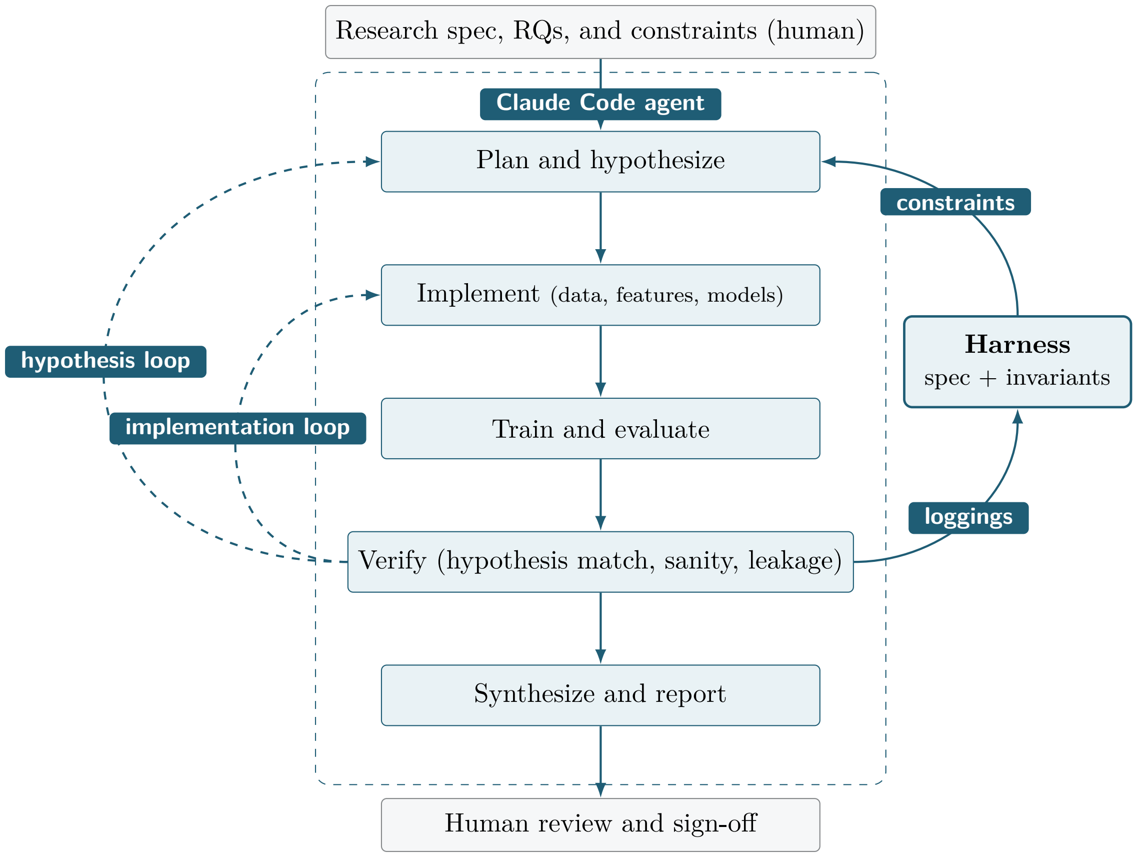

Figure 1: AutoResearch execution pipeline. The agent owns the steps inside the dashed boundary; humans handle the spec and final review. An inner *implementation loop* (verify → implement) handles code, data, or numerical issues; an outer *hypothesis loop* (verify → plan) handles cases in which results indicate the hypotheses themselves require revision. Each loop iteration that fails verification deposits a failure-mode entry into the *harness*, which combines the original human spec with accumulated invariants and is consulted at every planning step.

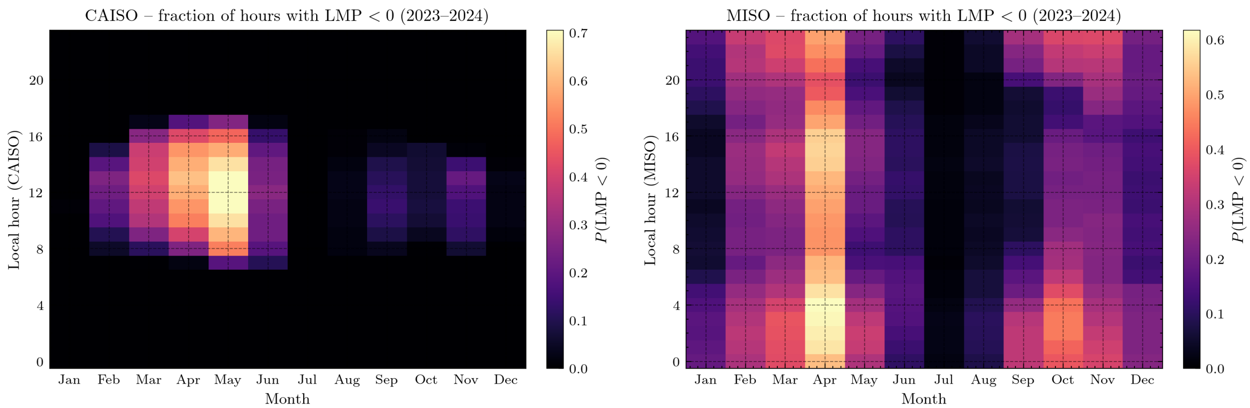

Figure 2: The two regimes. *Left*: CAISO. Negative LMPs concentrate sharply between 10:00 and 15:00 PT during the spring shoulder (February–May), peaking near 70% of hours in April–May midday. Outside that window, negatives are essentially absent (dark cells). The pattern is classic solar-driven oversupply. *Right*: MISO wind-belt nodes. Negatives are spread across all 24 hours of the day, with seasonal peaks in spring and autumn, and a near-zero July–August trough. There is no diurnal concentration. The pattern is wind-and-congestion-driven.

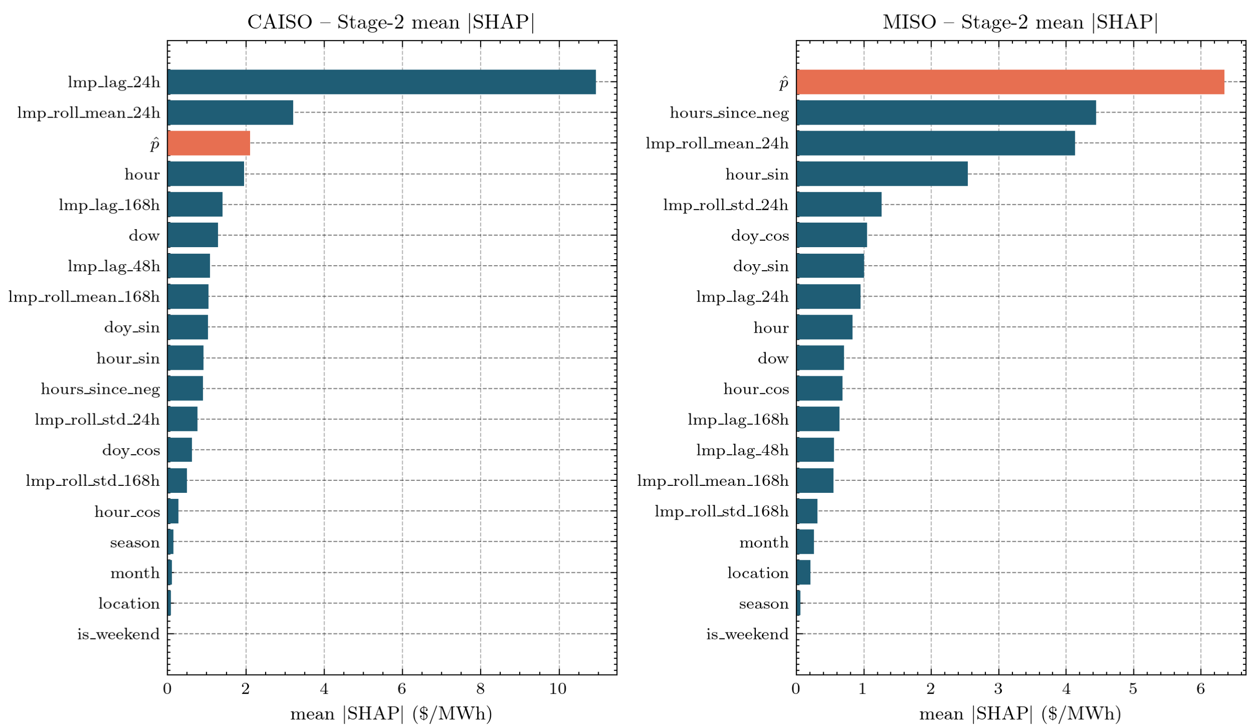

Figure 3: Mean absolute SHAP values for Stage-2 features. In MISO, the Stage-1 probability `p_hat` is the dominant feature.

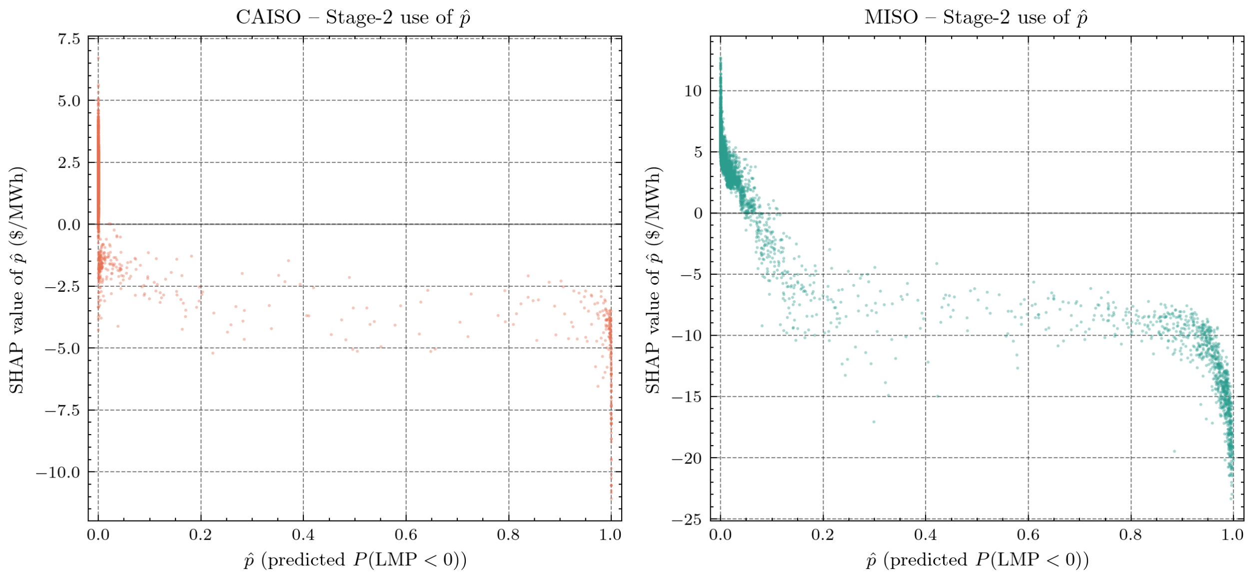

Figure 4: Effect of the Stage-1 probability `p_hat` on Stage-2 predictions. Each point is an hourly observation. In CAISO, `p_hat` is concentrated near zero and one. In MISO, it varies continuously across the interval.

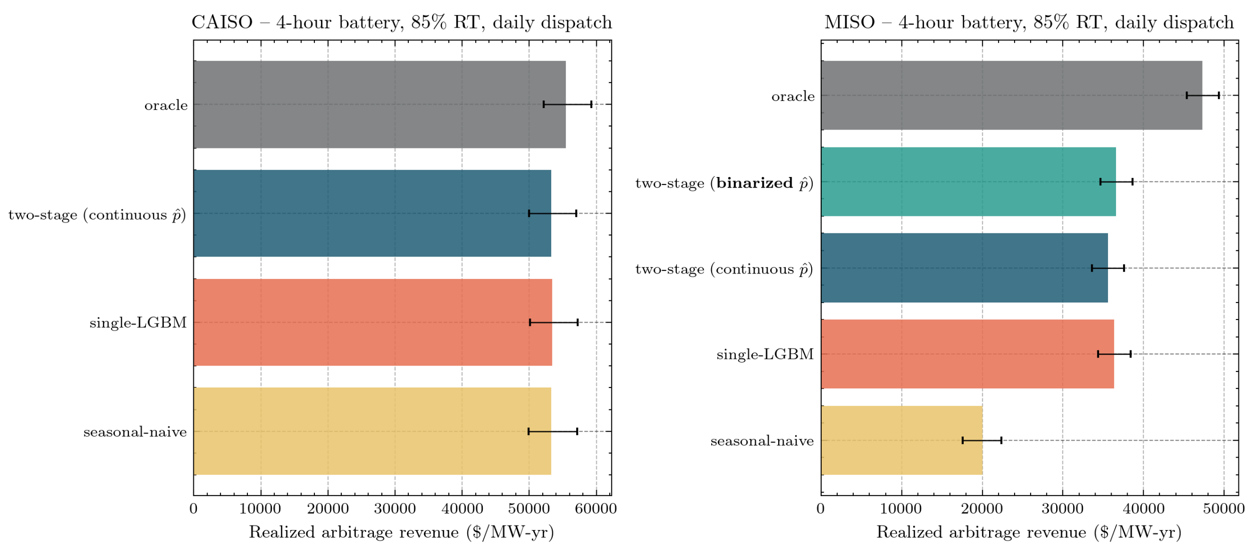

Figure 5: Annualized realized battery-arbitrage revenue by market and forecasting model. Error bars are bootstrap 95% confidence intervals.

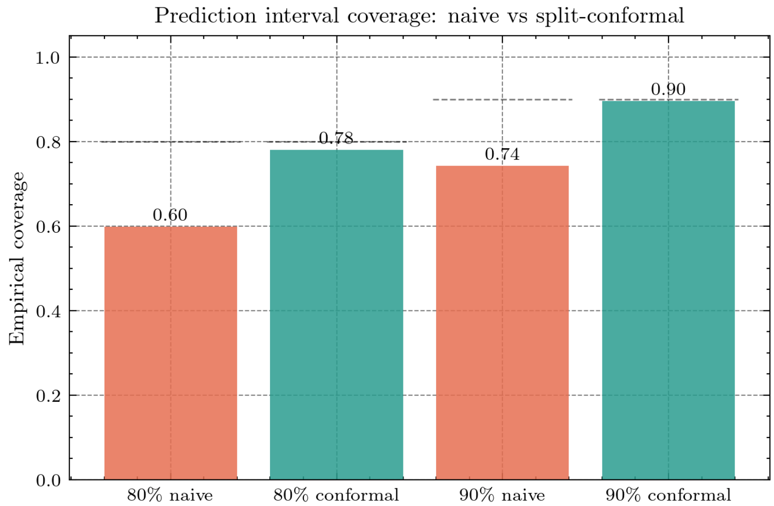

Figure 6: Empirical coverage of raw and calibrated prediction intervals. Dashed lines mark the intended coverage.

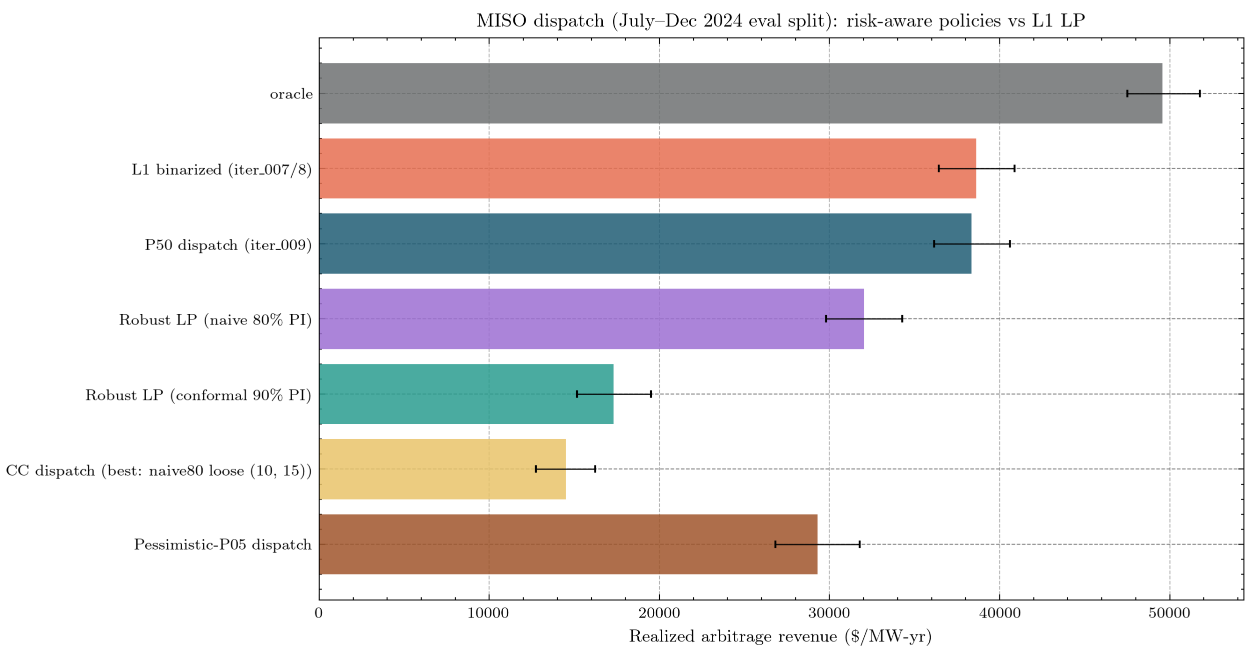

Figure 7: MISO battery-arbitrage revenue under the policies defined above. Oracle revenue is included as an upper bound.

A Mathematical specification of the two-stage forecaster

Let D index delivery days and $h \in \{0, \ldots, 23\}$ index hours of the day. At gate-closure of day $D-1$, the operator forecasts the 24 hourly LMPs $\{\mathrm{LMP}_{D,h}\}$ for delivery day D. The negative-event indicator is

$$Y_{D,h} \; = \; \mathbf{1}\{\mathrm{LMP}_{D,h} < 0\},$$

and the regression target is $\mathrm{LMP}_{D,h} \in \mathbb{R}$. The feature vector $x_{D,h} \in \mathbb{R}^p$ contains only quantities observable by gate closure.

Stage 1 — event detection (LightGBM classifier)

Stage 1 fits a classifier g_{θ_1} for the conditional probability of a negative event:

$$\hat{p}_{D,h} \; = \; \widehat{\Pr}(Y_{D,h} = 1 \mid x_{D,h}) \; = \; g_{\theta_1}(x_{D,h}).$$

We train under *weighted binary cross-entropy* with inverse-frequency class weights:

$$\mathcal{L}_1(\theta_1) \; = \; -\sum_{(D,h)\in\mathcal{T}} \big[w_1 Y_{D,h} \log \hat{p}_{D,h} \; + \; w_0 (1 - Y_{D,h}) \log(1 - \hat{p}_{D,h}) \big],$$

$$w_c \; = \; \frac{n}{2n_c}, \qquad c \in \{0, 1\}, \qquad n_c = \#\{(D,h) \in \mathcal{T} : Y_{D,h} = c\}.$$

The classifier outputs a probability $\hat{p}_{D,h}$ rather than a binary prediction. For classification metrics, we convert this probability into a predicted label, $\hat{Y}_{D,h} = \mathbf{1}\{\hat{p}_{D,h} \geq \tau^\star\}$. This threshold is used only for classification evaluation; it does not affect model training or the Stage-2 regression input. We choose $\tau^\star$ on validation to maximize F_1 subject to a recall floor of 0.70:

$$\tau^\star \; = \; \arg\max_{\tau \in [0,1]} F_1(\tau) \quad \text{s.t.} \quad \mathrm{Recall}(\tau) \geq 0.70.$$

Stage 2 — magnitude estimation (LightGBM regressor)

Stage 2 fits a regressor for the LMP value, with the Stage-1 probability as an extra feature:

$$\widehat{\mathrm{LMP}}_{D,h} \; = \; f_{\theta_2}(x_{D,h}, \, \hat{p}_{D,h}).$$

Training uses L_1 (MAE) loss:

$$\mathcal{L}_2(\theta_2) \; = \; \sum_{(D,h)\in\mathcal{T}'} \big| \mathrm{LMP}_{D,h} - \widehat{\mathrm{LMP}}_{D,h} \big|.$$

Out-of-fold Construction

Feeding $\hat{p}_{D,h}$ from a Stage-1 model fit on the same (D, h) would introduce overfit information into Stage 2. We construct $\hat{p}$ as an *out-of-fold* prediction via `TimeSeriesSplit`: training is partitioned into $K = 5$ chronological folds, and for each fold the Stage-1 prediction on the held-out segment uses a Stage-1 model fit only on prior segments. The first $1/(K + 1)$ of the training set is dropped from Stage-2 training because no out-of-fold prediction exists for it. On data outside training, Stage 1 is refit on all training data and used directly.

Binarized Variant

In the binarized variant, the probability fed to Stage 2 is thresholded at $\tau_b = 0.5$ at both training and prediction time:

$$\hat{p}^{(\mathrm{bin})}_{D,h} \; = \; \mathbf{1}\{\hat{p}_{D,h} \geq \tau_b\} \in \{0, 1\}.$$

This is the only structural difference between the continuous-probability and binarized two-stage architectures.

Baselines

- **Seasonal naive**: $\hat{\text{LMP}}_{D,h} = \text{LMP}_{D-1,h}$ (yesterday at same hour); also reported with $D-7$ (last week at same hour).
- **End-to-end LightGBM**: identical to Stage 2 but without the Stage-1 `p_hat` feature; trained directly on $(x_{D,h}, \text{LMP}_{D,h})$ under L_1 loss.

B Evaluation metrics

Classification metrics

Let $\hat{Y} = \mathbf{1}\{\hat{p} \geq \tau^\star\}$. Define

$$P = \frac{\text{TP}}{\text{TP}+\text{FP}}, \quad R = \frac{\text{TP}}{\text{TP}+\text{FN}}, \quad F_1 = \frac{2PR}{P+R}.$$

PR-AUC is the area under the precision-recall curve, computed as the average precision over all thresholds. It is preferred over ROC-AUC for imbalanced problems because it does not reward confidently predicting the (easy) negative class. **Brier score** is the mean squared error of the probability: $\text{Brier} = \frac{1}{n}\sum(\hat{p}-Y)^2$.

Regression metrics

$$\text{MAE} = \frac{1}{n}\sum_{(D,h)} |\text{LMP}_{D,h} - \hat{\text{LMP}}_{D,h}|, \quad \text{RMSE} = \sqrt{\frac{1}{n}\sum_{(D,h)} (\text{LMP}_{D,h} - \hat{\text{LMP}}_{D,h})^2}.$$

MAE-on-negatives restricts the sum to $\{(D,h) : Y_{D,h} = 1\}$:

$$\text{MAE-neg} \;=\; \frac{1}{n_{\text{neg}}}\sum_{Y=1} |\text{LMP} - \hat{\text{LMP}}|.$$

Bootstrap 95% confidence intervals use 2,000 resamples of the per-observation absolute-error series.

Pinball loss (quantile regression)

For a quantile $q \in (0,1)$ and prediction $\hat{y}_q$:

$$\rho_q(y, \hat{y}_q) \;=\; (y - \hat{y}_q)\big(q - \mathbf{1}\{y < \hat{y}_q\}\big).$$

Mean pinball loss at $q = 0.5$ equals half the MAE.

Diebold–Mariano test with Harvey–Leybourne–Newbold correction

The standard EPF pairwise comparison [1]. Given per-observation losses $L_A(t)$ and $L_B(t)$, define the loss differential $d_t = L_A(t) - L_B(t)$. The null is $\mathbb{E}[d_t] = 0$. The long-run variance is estimated with a Bartlett kernel up to lag $h-1$:

$$\hat{\sigma}^2 \;=\; \hat{\gamma}_0 + 2\sum_{k=1}^{h-1}\Big(1 - \tfrac{k}{h}\Big)\hat{\gamma}_k, \qquad \hat{\gamma}_k = \tfrac{1}{n}\sum_{t=k+1}^{n}(d_t - \bar{d})(d_{t-k} - \bar{d}).$$

Raw DM statistic: $\text{DM} = \bar{d}/\sqrt{\hat{\sigma}^2/n}$. The HLN small-sample correction is

$$\text{DM}^\star \;=\; \text{DM}\cdot\sqrt{k_{\text{HLN}}}, \qquad k_{\text{HLN}} \;=\; \tfrac{n+1-2h+h(h-1)/n}{n}.$$

Under the null, $\text{DM}^\star \sim t_{n-1}$ for two-sided p-values. We additionally check stationarity of $\{d_t\}$ via the augmented Dickey–Fuller test ($p_{\text{ADF}} < 0.05$); reported p-values are valid only when ADF rejects unit-root.

C Data details

Sources

LMPs are pulled through `gridstatus` [21], an open-source Python wrapper around CAISO OASIS [18, 19] and MISO Data Exchange [20]. Both interfaces are public; no credentials were required for the queries used here. ERCOT was excluded because its public-data API requires registration and a subscription key that were not provisioned.

Empirical date ranges and splits

ISO	Train	Val	Test	Hours total
CAISO	2023-03-01 → 2024-02-29	2024-03-01 → 2024-05-31	2024-06-01 → 2024-12-31	48,448
MISO	2023-01-01 → 2024-02-29	2024-03-01 → 2024-05-31	2024-06-01 → 2024-12-31	69,504

Pre-2023 CAISO data are not retrievable from the OASIS endpoint for the selected trading-hub node list.

Locations

CAISO (3 trading hubs): TH_NP15_GEN-APND (Northern California), TH_SP15_GEN-APND (Southern California), TH_ZP26_GEN-APND (Central California).

MISO (4 wind-belt nodes): chosen empirically from 864 MISO day-ahead location IDs by selecting nodes with >10% negative-LMP rate over the full 2023–2024 cache. The named MISO "hubs" (INDIANA.HUB, ILLINOIS.HUB, etc.) have ∼0% negative rate due to aggregation washout and are unsuitable as classification targets.

- MEC.OBRIEN.MVP (Iowa, MidAmerican Energy; 18.6% negative rate)
- MDU.ELLNDL.MVP (North Dakota, Montana-Dakota Utilities; 29.4%)
- NSP.HCPD_1.AZ (Minnesota, Northern States Power; 16.7%)
- ALTW.LEDYD.MVP (Iowa, Alliant West; 16.0%)

Feature engineering

Features include per-location calendar variables (hour, day of week, month, season, sinusoidal hour, and sinusoidal day of year); price lags at 24h, 48h, and 168h; 24h and 168h rolling means and standard deviations; a past-only `hours_since_neg` persistence counter; and an integer location encoding. All lagged features are observable before day-ahead gate closure.

D Battery arbitrage LP

Battery: 4 MWh capacity, 1 MW maximum charge/discharge rate, 85% round-trip efficiency, daily state-of-charge (SoC) reset to 0. Let $\eta_{\text{chg}} = \eta_{\text{dch}} = \sqrt{0.85} \approx 0.922$.

LP per day (24 hours, indexed by t):

$$\begin{aligned} \max_{c,d,\text{SoC}} \quad & \sum_{t=0}^{23} (d_t - c_t) \cdot \hat{p}_t \\ \text{s.t.} \quad & \text{SoC}_t = \text{SoC}_{t-1} + \eta_{\text{chg}} c_t - d_t/\eta_{\text{dch}}, \\ & 0 \le \text{SoC}_t \le C, \quad 0 \le c_t, d_t \le P_{\max}, \quad \text{SoC}_0 = 0. \end{aligned}$$

where $\hat{p}_t$ is the *forecast* price for hour t. The dispatch schedule $\{c_t, d_t\}$ is then settled at the *actual* realized prices to give realized revenue

$$R_{\text{day}}^{\text{realized}} = \sum_{t=0}^{23} (d_t - c_t) \cdot \text{LMP}_t^{\text{actual}}.$$

Annualized revenue per MW: $\mathrm{mean}_{\mathrm{days}}(R_{\mathrm{day}}^{\mathrm{realized}}) \times 365$. Pairwise comparisons across forecasters use paired-bootstrap (2,000 resamples) on the per-(location, day) realized-revenue differential.

The LP is solved by `scipy.optimize.linprog` (HiGHS solver). On 1,484 location-days (CAISO + MISO test set), the full backtest runs in under a minute.

E SHAP attribution

We use `shap.TreeExplainer` on the LightGBM Stage-2 models. For interpretation, Stage 2 is refit on the entire pre-test data (train + validation) for each ISO and SHAP values are computed on a random sample of 5,000 test-period observations per ISO. The mean absolute SHAP value of a feature is reported as its importance; the SHAP value as a function of the feature value (the *dependence plot*) is the diagnostic used to identify the bimodal-vs-continuous regime contrast described in §5.3.

F Quantile regression and split-conformal calibration

Quantile regression

Stage 2 is retrained with LightGBM's quantile objective at $q \in \{0.05, 0.10, 0.50, 0.90, 0.95\}$ to produce five conditional-quantile forecasts per observation. The architecture is otherwise identical to the binarized two-stage model.

Split-conformal prediction (CQR)

Following [1] (for the EPF analogy) and the conformal-prediction literature, we apply Conformalized Quantile Regression (CQR). Let α be the target miscoverage rate (so $1-\alpha$ is the target coverage). Define the nonconformity scores on a calibration set $\{(x_i, y_i)\}_{i=1}^{n_{\mathrm{cal}}}$:

$$E_i \;=\; \max\big(\hat{y}_{\alpha/2}(x_i) - y_i,\;\; y_i - \hat{y}_{1-\alpha/2}(x_i)\big).$$

The conformal offset $\widehat{Q}$ is the $\lceil (n_{\mathrm{cal}}+1)(1-\alpha) \rceil / n_{\mathrm{cal}}$-th empirical quantile of $\{E_i\}$. For a test point, the calibrated interval is

$$\big[\,\hat{y}_{\alpha/2}(x_{\mathrm{test}}) - \widehat{Q},\;\; \hat{y}_{1-\alpha/2}(x_{\mathrm{test}}) + \widehat{Q}\,\big],$$

and has provable marginal coverage $\geq 1-\alpha$ under data exchangeability. For MISO the calibration set is the first 30 days of test (June 2024; $n_{\mathrm{cal}} = 2{,}880$); evaluation is the remaining 6 months (July–December 2024; $n_{\mathrm{eval}} = 17{,}664$).

G Risk-aware dispatch policies

The four risk-aware policies tested in §5.6:

- **Chance-constrained (CC) trade mask.** Run the standard LP using the P50 forecast, but force $c_t = 0$ unless the calibrated upper bound is below a buy threshold, and force $d_t = 0$ unless the calibrated lower bound is above a sell threshold. We swept across 6 (interval, threshold) configurations.
- **Robust LP.** Replace the objective with $\max \sum_t d_t \cdot \mathrm{lower}_t - \sum_t c_t \cdot \mathrm{upper}_t$ (assume the worst-case price within the interval for both directions); tested with both the conformal 90% and naive 80% bounds.
- **Pessimistic-P05 forecast.** Run the standard LP with the lower-quantile (P05 conformal) prediction as the price input. A pure point-forecast heuristic that is risk-averse by construction.
- **P50 baseline.** Run the standard LP with the conditional median (quantile-trained $q = 0.5$). Mathematically equivalent to the L1 LP; included as a sanity check.

None of the four policies outperforms the L1 LP baseline on the evaluation split.

H Extended limitations

1. **Time range.** The `gridstatus` OASIS endpoint returns no CAISO PRC_LMP data for the selected trading-hub node list prior to 2023-02-09. The empirical split is therefore 2023-03 / 2024-03–05 / 2024-06–12 for CAISO and 2023-01 / 2024-03–05 / 2024-06–12 for MISO.
2. **ERCOT excluded.** The ERCOT Public Data API [24] requires registration, a subscription key, and a bearer token that were unavailable for this study. A third ISO would provide an additional test of the binarized-`p_hat` architecture.
3. **Weather features excluded.** The NSRDB and WIND Toolkit require NREL API credentials that were unavailable for this study.
4. **LEAR baseline excluded.** The Lago et al. 2021 LEAR model via `epftoolbox` is designed around European EPF dataset formats. Adapting it to CAISO and MISO is left for future work.
5. **Battery retraining cadence.** We use eight 28-day rolling-origin windows over the seven-month test period. The cadence is shared by the compared models, but absolute MAE values may differ under more frequent retraining.
6. **Battery: daily SoC reset, single configuration.** A real operator carries SoC across days and might solve a 7-day LP; relaxing this would raise revenue across all models proportionally. Configurations other than 4-hour (e.g., 2-hour or 8-hour batteries) would shift absolute numbers but probably not the ranking.
7. **No cycling cost.** Real batteries amortize per-cycle degradation. Adding a cycling cost dampens all numbers but does not change the ranking.
8. **Risk-aware dispatch tested under only one decision problem.** The failure of risk-aware policies to beat the L1 LP is consistent with linear-utility decision theory in arbitrage. Future work should evaluate decision problems with asymmetric losses, such as reliability contracts or ancillary services.